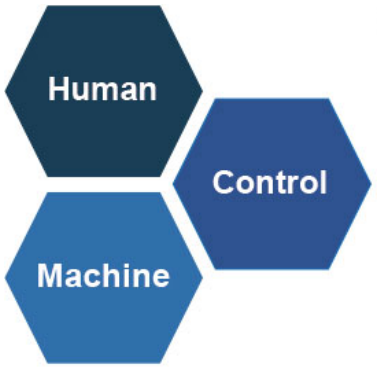
2026.03.05 일일 보고
학습 내용 정리

공민수 석사과정생

Motion Control Lab



We are dedicated to helping people find better ability in motions through various technologies of robotics and control engineering.
Four keywords of our research are:
Control, Actuator, Mobility and Muscle.

Knowledge and technologies on human,machine and control are converged to achieve our goals

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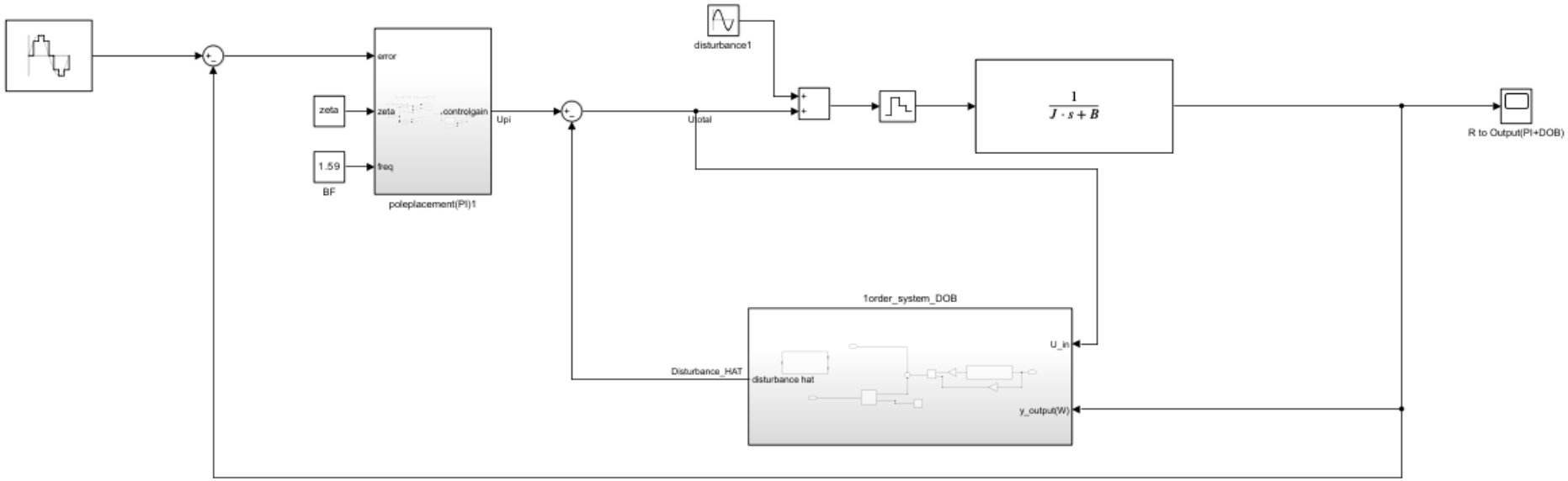
- **Pole Placement PI & DOB**
  - Tracking Performance with Bode plot
  - Disturbance Estimation Performance

- **Adding Feed Forward Control**
  - Change of Control Input & Tracking Performance

- **Adding the nominal values of maxon motor**
  - Effect of Saturation

Pole Placement PI & DOB 구현

- **Velocity Control Block Diagram with PI & DOB**



- **Arbitrary Constant Parameters**

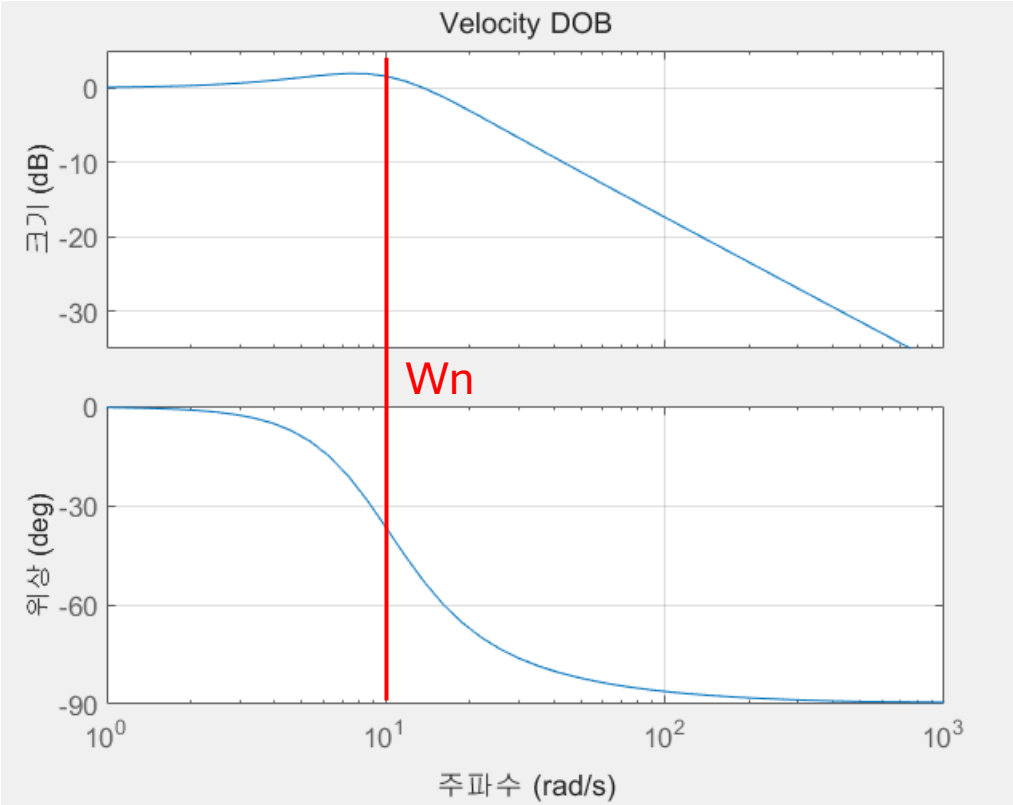
J	B	zeta	wn	Bandwidth	Magnitude Reference	Magnitude Disturbance	LPF Cutoff Fq
1[kg · m^2]	0.5[N · m · s]	0.7	10[rad/s]	1.59[Hz]	10[rad/s]	10[N · m]	10[N · m]

- **Varying Parameters**
  - Frequency of Reference
  - Frequency of Disturbance

Tracking Performance with Bode plot

- **Approach**
  - Relationship between Reference frequency and PI Bandwidth

- **Analysis Method**
  - Tracking Performance Observation by Adjusting Input Frequency Based on Bode Plot Gain and Phase

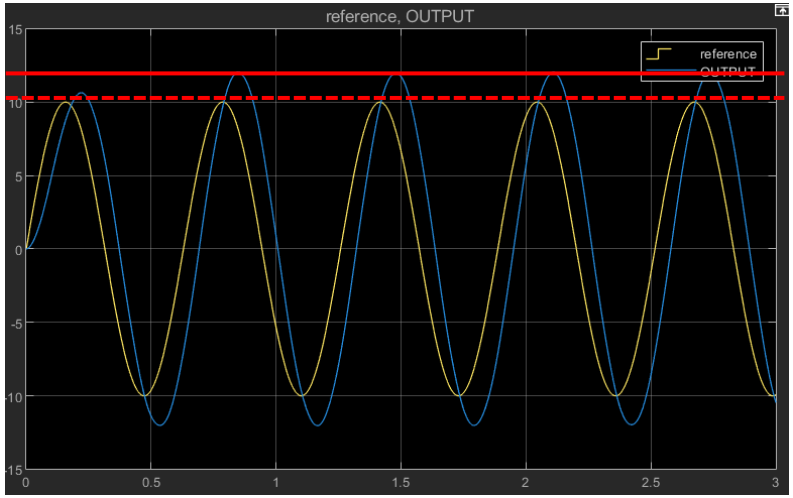


- **Gain = 1.1995**
- **Magnitude = 1.58 [dB]**
- **Phase = -36.57 [deg]**

Tracking Performance with Bode plot

- **Results**
  - Frequency of reference = Bandwidth

Ref
Output

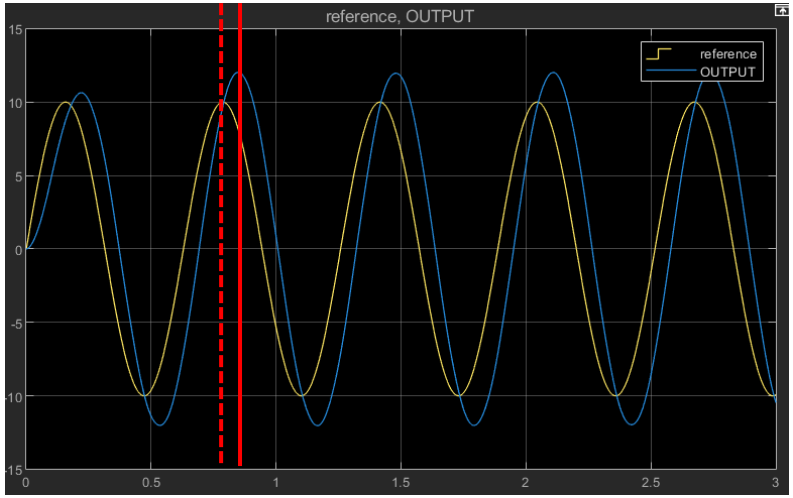


W[rad/s]

Time [s]

- **Output Peak = 12.02 [rad/s]**
- **Reference peak = 10 [rad/s]**

- **Gain = 1.202**



W[rad/s]

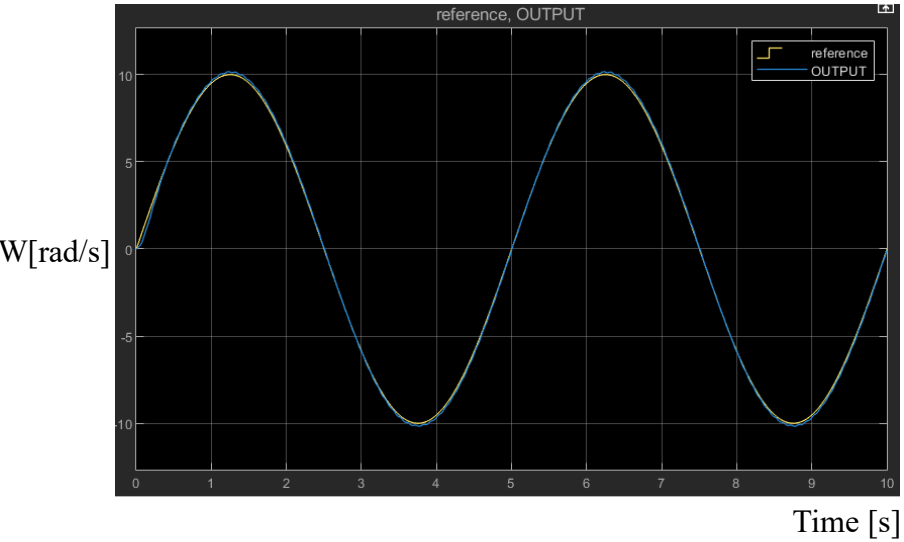
Time [s]

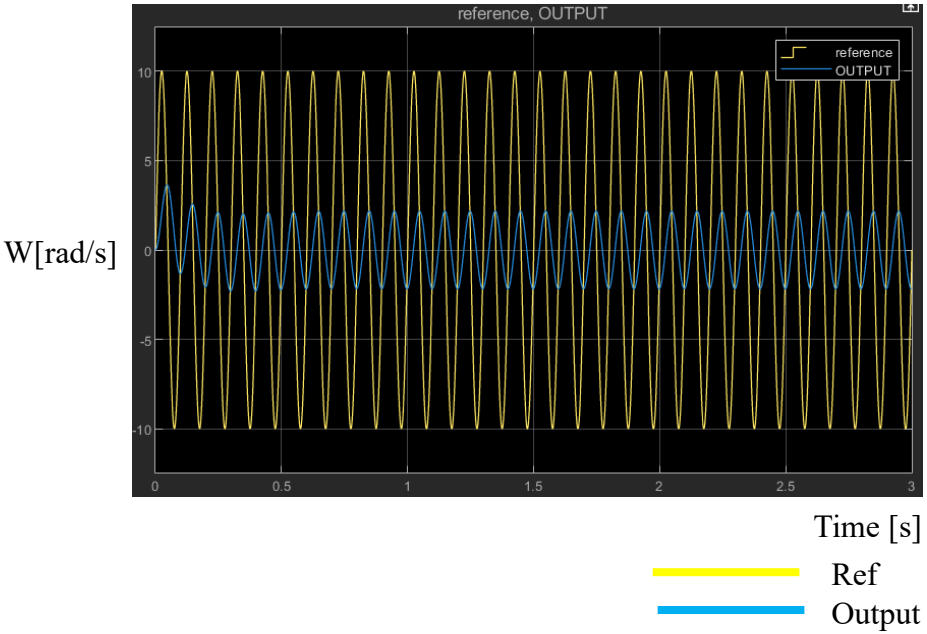
- **Phase Delay = 0.061 [s]**
- **Frequency= 1.59 [Hz]**

- **Phase = 34.91 [deg]**

Tracking Performance with Bode plot

- **Results**
  - Frequency of reference << Bandwidth
  - Frequency of reference > >Bandwidth





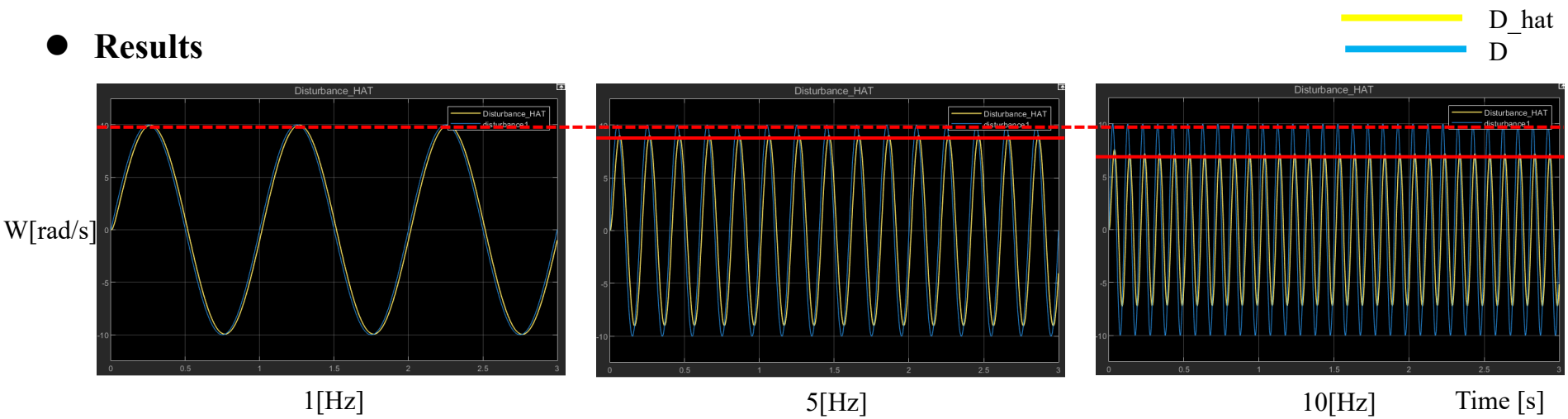
- **Conclusions**
  - The bandwidth in pole placement can be used to tune the frequency range for tracking.

Disturbance Estimation Performance

- **Approach**
  - Relationship between Reference frequency and PI Bandwidth

- **Analysis Method**
  - Estimation Performance by Adjusting Disturbance Frequency Based on 1st LPF

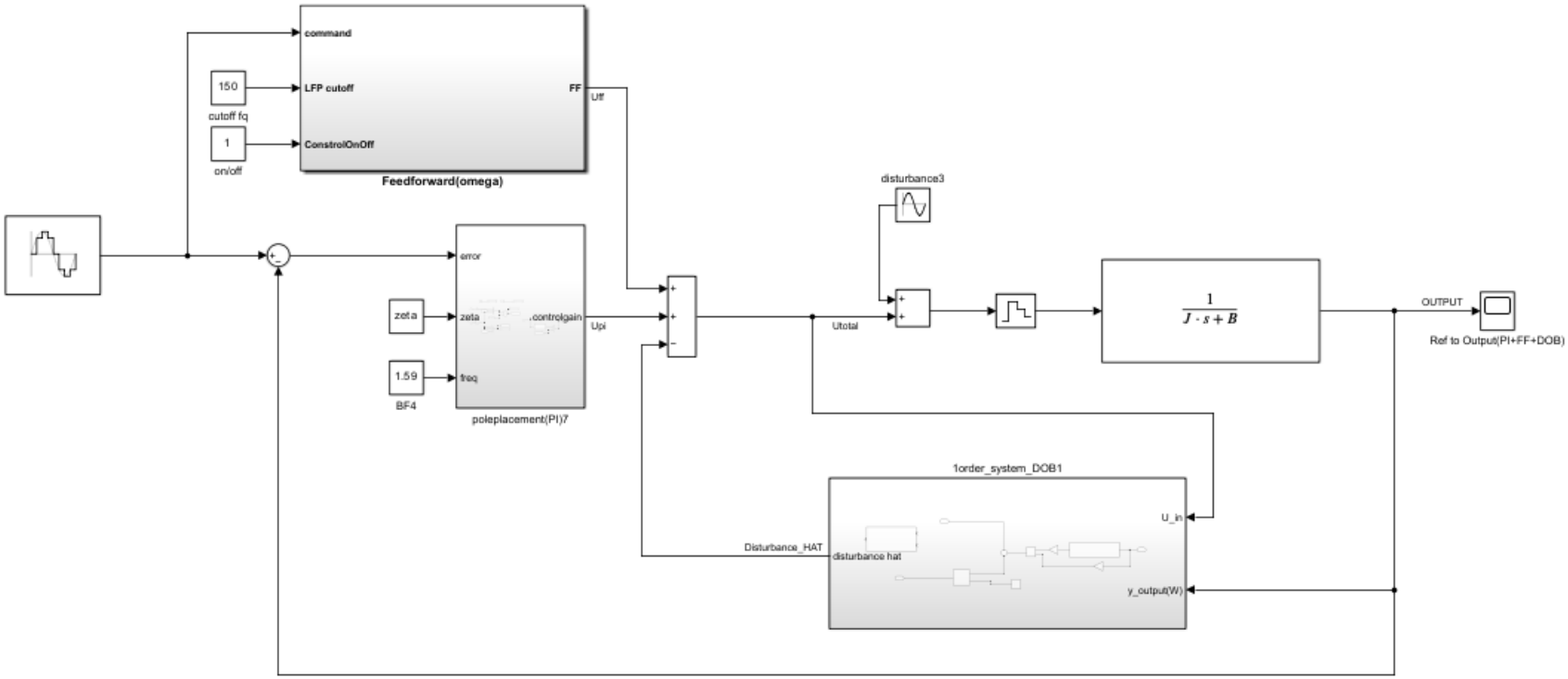
- **Results**



- **Conclusion**
  - The Disturbance estimation performance is influenced by the cutoff frequency of the LPF

Adding Feed Forward Control

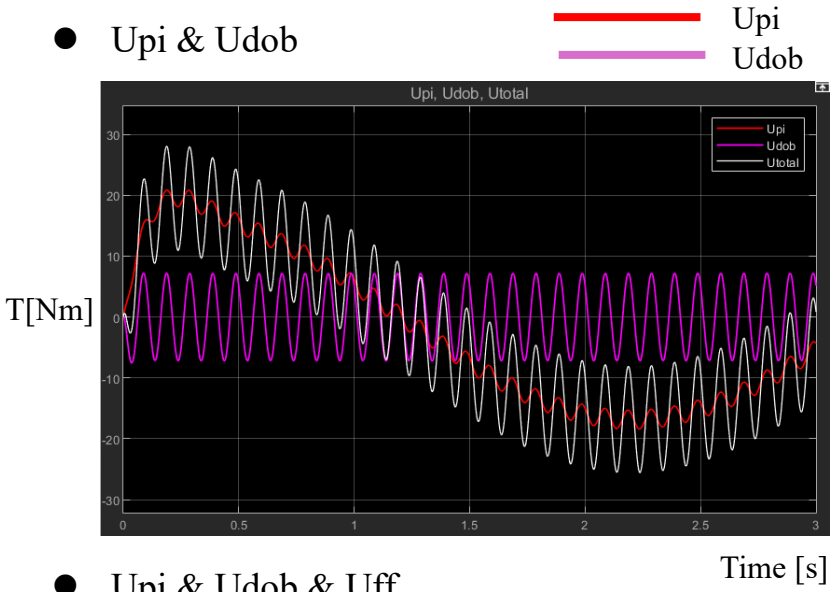
- **Velocity Control Block Diagram with PI, DOB & Feed Forward**



- **Parameters is the Same with 'Velocity output with PI & DOB' and Feed Forward Cutoff Frquency is 150 [Hz]**

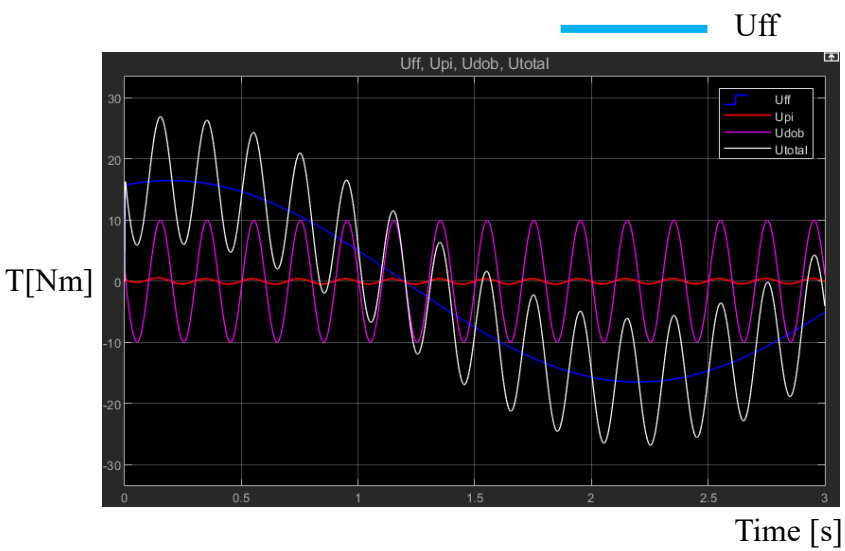
Change of Control Input & Tracking Performance

- **Change of Control Input Upi, Udob, Uff respectively**
  - Upi & Udob



- **-20 < Upi < 20** [N · m]
- **-10 < Udob < 10** [N · m]
- **-30 < Utotal < 30** [N · m]

  - Upi & Udob & Uff

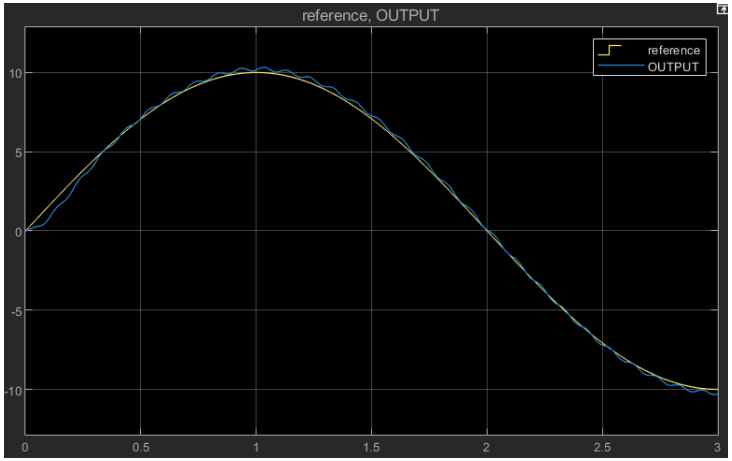


- **-2 < Upi < 2** [N · m]
- **-10 < Udob < 10** [N · m]
- **-20 < Uff < 20** [N · m]
- **-30 < Utotal < 30** [N · m]

Change of Control Input & Tracking Performance

- **Change of the Tracking Performance**
  - Upi & Udob

Ref
Output



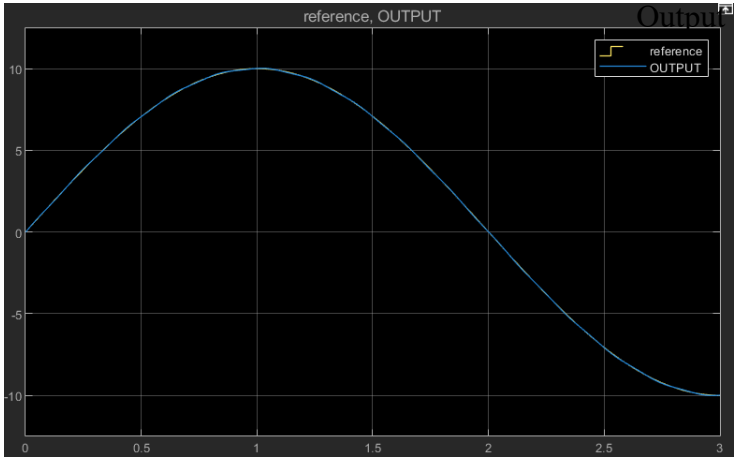
W[rad/s]

Time [s]

- **Phase lag and Tracking Error**

  - Upi & Udob & Uff

Ref
Output



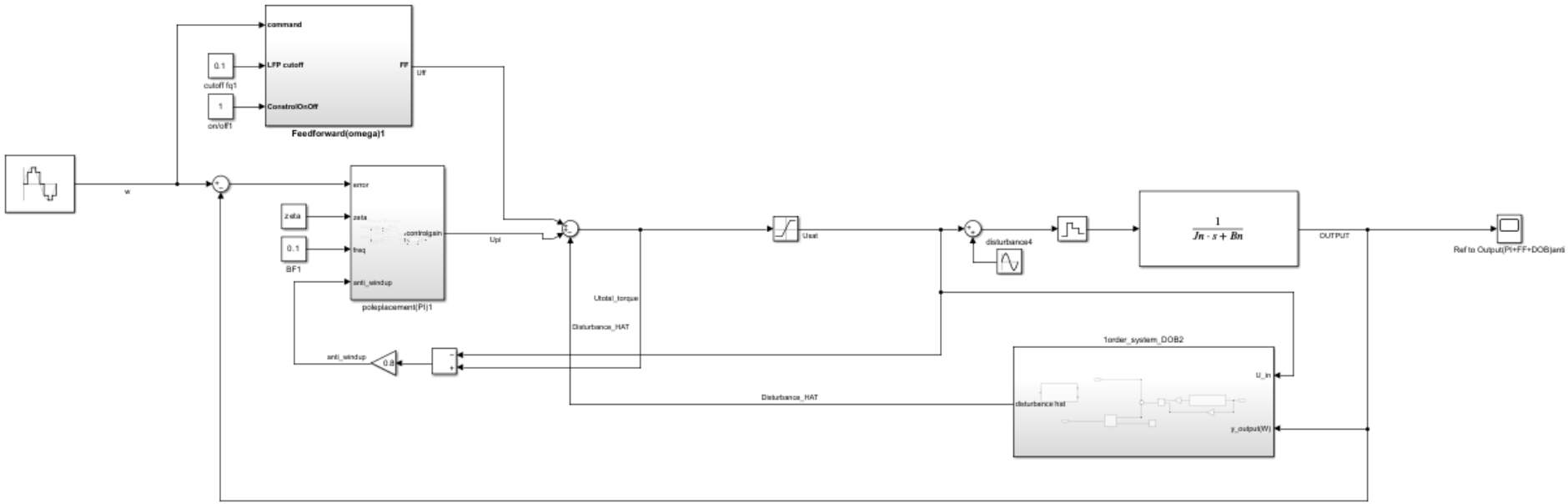
W[rad/s]

Time [s]

- **Good Error and response performance**

Adding the nominal values of maxon motor

- **Velocity Control Block Diagram with PI, DOB & Feed Forward (Maxon motor)**

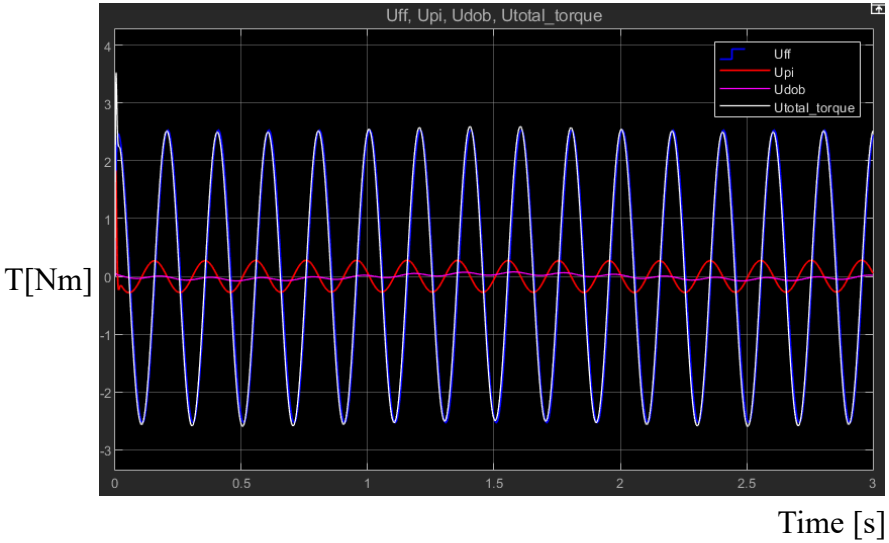


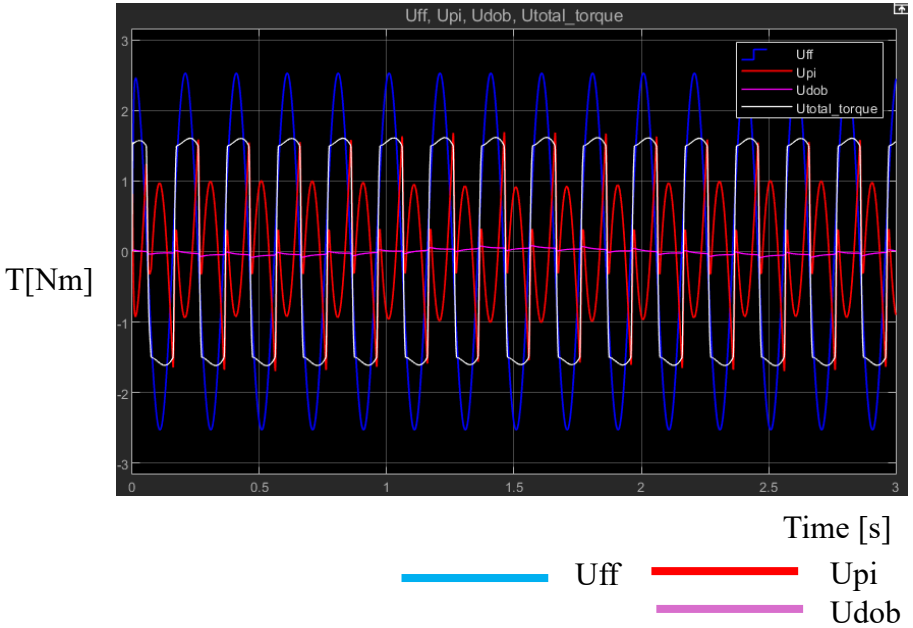
- **Nominal Parameters**

Jn	Bn	Saturation(Rated Torque)
$5.1 \times 10^{-4}[\mathrm{kg} \cdot \mathrm{m}^2]$	$3.07 \times 10^{-4}[\mathrm{N} \cdot \mathrm{m} \cdot \mathrm{s}]$	$1.56\ [\mathrm{N} \cdot \mathrm{m}]$

Effect of Saturation

- **Change of Control input**
  - Ideal Case
  - Saturation Case

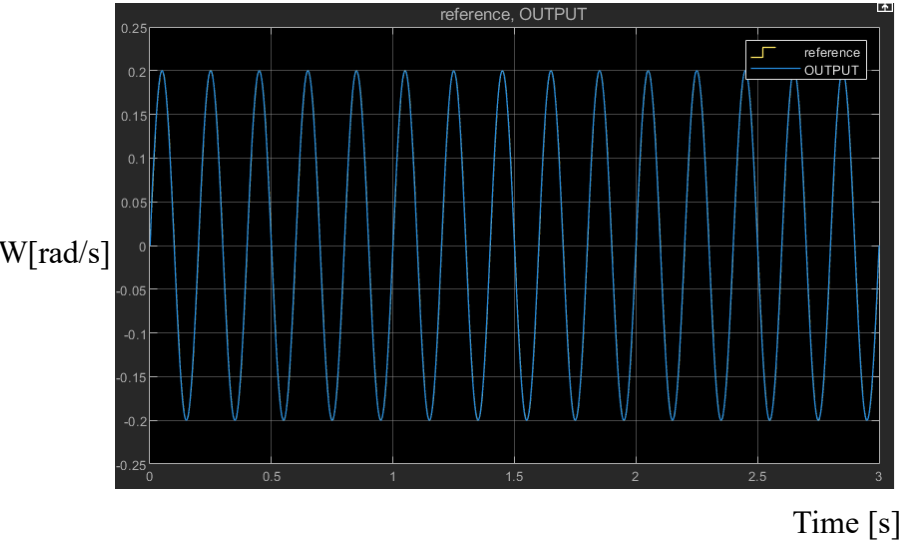


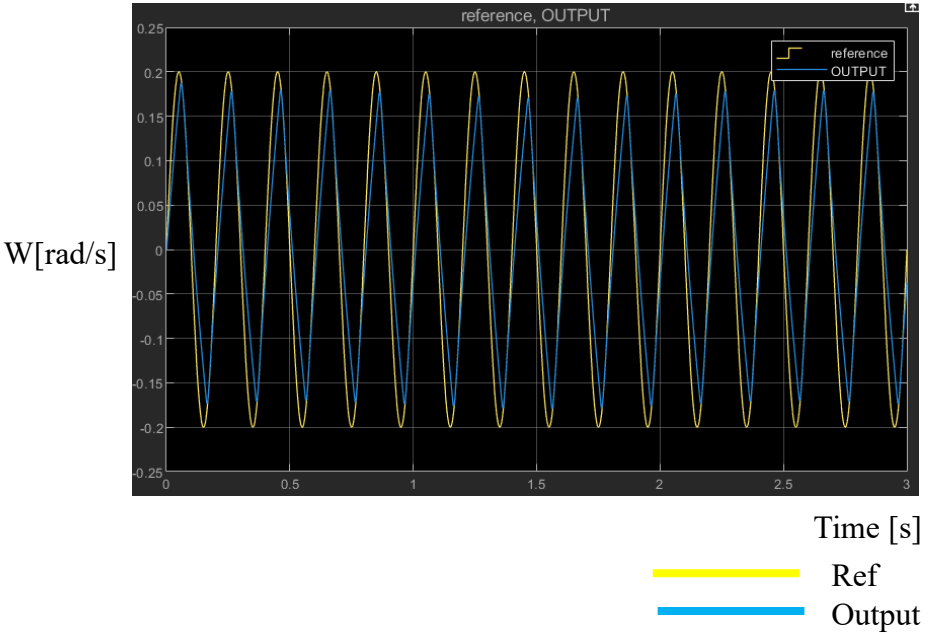


- **Analysis**
  - Saturation case's Upi is bigger than Ideal Case
  - Saturation case's Udob becomes irregular than Ideal Case

Effect of Saturation

- **Reference Tracking Performance**
  - Ideal Case
  - Saturation Case





- **Analysis**
  - Phase Lag, Magnitude Decrease
- **Conclusion**
  - Actuator saturation leads to error accumulation, which increases the integrator output $U_{pi}$and degrades the tracking performance.

TO DO

- Rover wheel을 통해 시뮬레이션으로 학습했던 내용들을 실습해본다.
- 특히 입력 주파수 대역과 Bandwidth, Cutoff Frequency를 고려하며 tracking튜닝을 진행한다.